Neyman-Pearson test

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Recap: hypothesis testing and power function

$$H_0 : \theta \in \Theta_0 \qquad H_A : \theta \in \Theta_1$$

Given observed data $X_1, ..., X_n$:

1. Calculate a test statistic $T_n = T(X_1, ..., X_n)$
2. Choose a rejection region $\mathcal{R} = \{(x_1, ..., x_n) : \text{ reject } H_0\}$
3. Reject H_0 if $(X_1, ..., X_n) \in \mathcal{R}$

The **power function** $\beta(\theta)$ is

$$\beta(\theta) = P_\theta((X_1, ..., X_n) \in \mathcal{R})$$

Goal: maximize power for $\theta \in \Theta_1$, subject to control of power for $\theta \in \Theta_0$

Neyman-Pearson test

Let $\mathbf{X} = (X_1, ..., X_n)$ be a sample from some distribution with probability function f and parameter θ. To test

$$H_0 : \theta = \theta_0 \qquad H_A : \theta = \theta_1,$$

the **Neyman-Pearson** test rejects H_0 when

$$\frac{L(\theta_1|\mathbf{X})}{L(\theta_0|\mathbf{X})} = \frac{f(X_1, ..., X_n|\theta_1)}{f(X_1, ..., X_n|\theta_0)} > k$$

where k is chosen so that $\beta(\theta_0) = \alpha$. (size α test)

Example

Suppose $X_1, \ldots, X_n \overset{indep.}{\sim} N(\theta, \sigma_i^2)$ where σ_i^2 are known, positive constants, and $\theta \in \mathbb{R}$ is unknown.

$$H_0 : \theta = 0 \qquad H_A : \theta = \theta_1 \quad \text{where } \theta_1 > 0$$

$$L(\theta|X) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left\{-\frac{1}{2\sigma_i^2}(X_i-\theta)^2\right\}$$

$$\Rightarrow \frac{L(\theta_1|X)}{L(0|X)} = \exp\left\{-\frac{1}{2}\left[\sum_{i=1}^{n}\frac{1}{\sigma_i^2}(X_i-\theta_1)^2 - \sum_{i=1}^{n}\frac{1}{\sigma_i^2}X_i^2\right]\right\}$$

$$= \exp\left\{-\frac{1}{2}\left[\sum_{i=1}^{n}\frac{1}{\sigma_i^2}(\cancel{X_i^2} - 2X_i\theta_1 + \theta_1^2 - \cancel{X_i^2})\right]\right\}$$

$$= \exp\left\{\frac{1}{2}\sum_{i=1}^{n}\frac{1}{\sigma_i^2}(2X_i\theta_1 - \theta_1^2)\right\}$$

$$\frac{L(\theta_1|X)}{L(0|X)} > K \iff \frac{1}{2}\sum_{i=1}^{n}\frac{1}{\sigma_i^2}(2X_i - \theta_1^2) > \log K$$

$$\iff \theta_1 \sum_{i=1}^{n}\frac{1}{\sigma_i^2}X_i - \frac{\theta_1^2}{2}\sum_{i=1}^{n}\frac{1}{\sigma_i^2} > \log K$$

$$\iff \sum_{i=1}^{n}\frac{X_i}{\sigma_i^2} > \underbrace{\frac{\log K + \frac{\theta_1^2}{2}\sum_{i=1}^{n}\frac{1}{\sigma_i^2}}{\theta_1}}_{= c}$$

Example

Suppose $X_1, \ldots, X_n \overset{indep.}{\sim} N(\theta, \sigma_i^2)$ where σ_i^2 are known, positive constants, and $\theta \in \mathbb{R}$ is unknown.

$$H_0 : \theta = 0 \qquad H_A : \theta = \theta_1 \quad \text{where } \theta_1 > 0$$

Neyman-Pearson test: reject when $\sum_{i=1}^{n} \frac{X_i}{\sigma_i^2} > c$ ← choose st $\beta(0) = \alpha$

Power: $X_i \sim N(\theta, \sigma_i^2) \Rightarrow \frac{X_i}{\sigma_i^2} \sim N\left(\frac{\theta}{\sigma_i^2}, \frac{1}{\sigma_i^2}\right)$

(independence)

$$\Rightarrow \sum_i \frac{X_i}{\sigma_i^2} \sim N\left(\theta \sum_{i=1}^{n} \frac{1}{\sigma_i^2}, \sum_{i=1}^{n} \frac{1}{\sigma_i^2}\right)$$

$$\beta(\theta) = P_\theta\left(\sum_{i=1}^{n} \frac{X_i}{\sigma_i^2} > c\right) =$$

$$P_\theta\left(\frac{\sum_{i=1}^{n} \frac{X_i}{\sigma_i^2} - \theta \sum_{i=1}^{n} \frac{1}{\sigma_i^2}}{\sqrt{\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}} > \frac{c - \theta \sum_{i=1}^{n} \frac{1}{\sigma_i^2}}{\sqrt{\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}}\right)$$

$N(0,1)$

$N(0,1)$ cdf

$$= 1 - \Phi\left(\frac{c}{\sqrt{\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}} - \theta\sqrt{\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}\right)$$

$$\beta(0) = 1 - \Phi\left(\frac{c}{\sqrt{\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}}\right) \overset{\text{set}}{=} \alpha \Rightarrow c = \Phi^{-1}(1-\alpha) \cdot \sqrt{\sum_{i=1}^{n} \frac{1}{\sigma_i^2}}$$

Example

Suppose $X_1, \ldots, X_n \overset{indep.}{\sim} N(\theta, \sigma_i^2)$ where σ_i^2 are known, positive constants, and $\theta \in \mathbb{R}$ is unknown.

$$H_0 : \theta = 0 \qquad H_A : \theta = \theta_1 \quad \text{where } \theta_1 > 0$$

Neyman-Pearson test: $\beta(\theta) = 1 - \Phi\left(\frac{c}{\sqrt{\sum_{i=1}^n \frac{1}{\sigma_i^2}}} - \theta\sqrt{\sum_{i=1}^n \frac{1}{\sigma_i^2}}\right)$

What if we just do a Wald test?

$$\sum_{i=1}^n X_i \sim N\left(n\theta, \sum_{i=1}^n \sigma_i^2\right) \Rightarrow \bar{X} \sim N\left(\theta, \frac{1}{n^2}\sum_{i=1}^n \sigma_i^2\right)$$

Wald test: reject when $\bar{X} > c_2$ (different from c above)

<u>Power</u>: $\beta^*(\theta) = P_\theta(\bar{X} > c_2)$

$$= P_\theta\left(\frac{\bar{X} - \theta}{\sqrt{\frac{1}{n^2}\sum_{i=1}^n \sigma_i^2}} > \frac{c_2 - \theta}{\sqrt{\frac{1}{n^2}\sum_{i=1}^n \sigma_i^2}}\right)$$

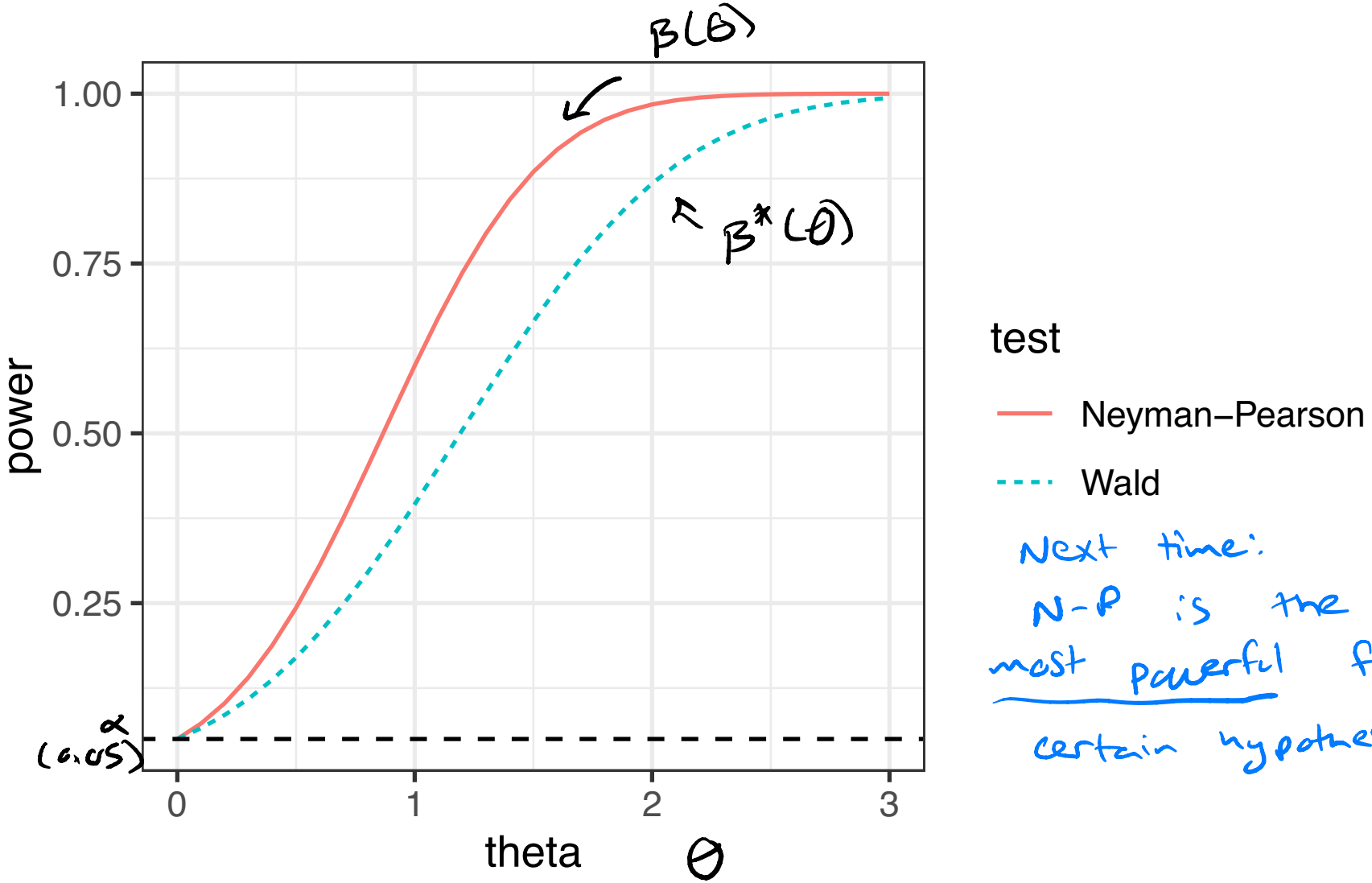
$$= 1 - \Phi\left(\frac{c_2 - \theta}{\sqrt{\frac{1}{n^2}\sum_{i=1}^n \sigma_i^2}}\right)$$

Example

$$X_1, \ldots, X_{20} \overset{indep.}{\sim} N(\theta, \sigma_i^2) \text{ where } \sigma_i^2 = i$$

$$H_0 : \theta = 0 \qquad H_A : \theta = \theta_1 \quad \text{where } \theta_1 > 0$$

$\alpha = 0.05$



Next time:
N-P is the most powerful for certain hypotheses

Example

Let $X_1, ..., X_n \overset{iid}{\sim} Exponential(\theta)$, with pdf $f(x|\theta) = \theta e^{-\theta x}$. We want to test

$$H_0 : \theta = \theta_0 \qquad H_A : \theta = \theta_1,$$

where $\theta_1 < \theta_0$. The Neyman-Pearson test rejects when

$$\frac{L(\theta_1|\mathbf{X})}{L(\theta_0|\mathbf{X})} > k.$$

1. Calculate $\dfrac{L(\theta_1|\mathbf{X})}{L(\theta_0|\mathbf{X})}$
2. Rearrange the ratio to show that $\dfrac{L(\theta_1|\mathbf{X})}{L(\theta_0|\mathbf{X})} > k$ if and only if $\sum_i X_i > c$ for some c
3. Using the fact that $\sum_i X_i \sim Gamma(n, \theta)$, find c such that $\beta(\theta_0) = \alpha$

Solution:

$$L(\theta \mid X) = \prod_{i=1}^{n} \theta e^{-\theta x_i} = \theta^n e^{-\theta \sum_i x_i}$$

$$\Rightarrow \frac{L(\theta_1 \mid X)}{L(\theta_0 \mid X)} = \left(\frac{\theta_1}{\theta_0}\right)^n e^{(\theta_0 - \theta_1) \sum_i x_i}$$

$\frac{L(\theta_1 \mid X)}{L(\theta_0 \mid X)}$ is an increasing function of $\sum_i X_i$ (since $\theta_0 - \theta_1 > 0$)

So, $\frac{L(\theta_1 \mid X)}{L(\theta_0 \mid X)} > k \iff \sum_i X_i > c$ for some c

$\sum_i X_i \sim \text{Gamma}(n, \theta)$ Let $F_{n,\theta}$ be cdf of $\text{Gamma}(n, \theta)$

$P_\theta(\sum_i X_i > c) = 1 - F_{n,\theta}(c) \Rightarrow \beta(\theta) = 1 - F_{n,\theta}(c)$

Now find c: want $\beta(\theta_0) = \alpha \Rightarrow 1 - F_{n,\theta_0}(c) = \alpha$

$\Rightarrow c = F_{n,\theta_0}^{-1}(1-\alpha)$ (upper α quantile of $\text{Gamma}(n, \theta_0)$)